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IS 492

CHECKPOINT 3 PRESENTATION

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Who Needs Sentinel & Why

Synthetic media is outpacing the tools designed to catch it.

THE PROBLEM

1. Deepfakes & NCEI at scale

AI-generated faces, cloned voices, and manipulated video are flooding platforms faster than humans can review.

2. Fragmented tooling

Analysts jump between detection APIs, EXIF viewers, and threat intel dashboards — no unified workflow.

3. High-stakes decisions, low context

Journalists, law enforcement, and trust & safety teams need explainable verdicts, not just a confidence score.

OUR SOLUTION

Sentinel unifies deepfake detection, media forensics, and AI analyst narratives into a single API-first platform.

Hive Moderation API
Deepfake confidence scoring

EXIF + VirusTotal
Media forensics & threat intel

Claude (Anthropic) / Groq (Llama 3)
Plain-language verdict narratives + Taxonomy Classification

Who uses Sentinel — and how?

Journalists & Researchers

USE CASE: Rapidly verify viral images/clips before publication

PAIN POINT: No time, no forensics background

SENTINEL: Instant verdict + plain-English narrative

Trust & Safety Analysts

USE CASE: Triage reported deepfakes and NCEI across large platforms

PAIN POINT: Volume too high for manual review

SENTINEL: Batch API, structured JSON, searchable archive

Law Enforcement / Legal Teams

USE CASE: Build case files with forensic-quality reports

PAIN POINT: Evidence needs to be exportable and auditable

SENTINEL: Export JSON, hash-based deduplication, 90-day retention

Security Operations

USE CASE: Monitor aggregate threat landscape via dashboard

PAIN POINT: No visibility into synthetic media trends

SENTINEL: Real-time stats, threat level charts, timeline view

Demo 2-Add Taxonomy Feature & AI models Selection (In Progress)

Reference from MIT AI Risk Repository

MIT AI Risk Initiative

- The **AI Risk Database** captures 1700+ risks extracted from 74 existing frameworks and classifications of AI risks
- The **Causal Taxonomy of AI Risks** classifies how, when, and why these risks occur
- The **Domain Taxonomy of AI Risks** classifies these risks into 7 domains and 24 subdomains (e.g., "False or misleading information")

Causal Taxonomy

Entity	Intent	Timing
AI: Due to a decision or action made by an AI system	**Intentional**: Due to an expected outcome from pursuing a goal	**Pre-deployment**: Before the AI is deployed
Human: Due to a decision or action made by humans	**Unintentional**: Due to an unexpected outcome from pursuing a goal	**Post-deployment**: After the AI model has been trained and deployed
Other: Due to some other reason or ambiguous	**Other**: Without clearly specifying the intentionality	**Other**: Without a clearly specified time of occurrence

https://airisk.mit.edu/

Demo 2-Add Taxonomy Feature & AI models Selection (In Progress)

Github branch - Taxonomy-AI-models

SENTINEL
DEEPFAKE INTELLIGENCE

Dashboard
Analyze
Reports

ANALYSIS RESULT

DEEPFAKE

CRITICAL THREAT

99% CONF

ID: 188764bb-a8e9-4d19-857d-006eaa05b9ae · 10092ms

// AI ANALYST NARRATIVE

THREAT INTELLIGENCE SUMMARY A high-confidence deepfake image (99.28% detection accuracy) has been identified in a 453KB PNG file with transparency layers, indicating sophisticated synthetic media generation likely designed for visual deception campaigns. Technical indicators include RGBA color mode with transparency support and stripped EXIF metadata suggesting deliberate preparation for distribution, while MITRE framework analysis points to potential use in phishing operations or influence campaigns targeting data manipulation. Immediate containment and forensic preservation are recommended, along with cross-referencing against known disinformation campaigns and implementing enhanced monitoring for related synthetic media artifacts.

Technical Metadata

MITRE ATT&CK MAPPING

T1566 - Phishing — Deepfake audio/video used in spear-phishing

T1565.001 - Stored Data Manipulation

Show the Causal Taxonomy result

MIT CAUSAL TAXONOMY — Claude (Anthropic)

ENTITY	INTENT	TIMING
Human	Intentional	Post-deployment

CLASSIFICATION CONFIDENCE 85%

The deepfake was created by humans using AI tools for malicious purposes (phishing/disinformation), indicating intentional misuse of deployed AI technology.

System Online

Model Selection

AI Model: Claude (Anthropic) | Llama 3 (Groq)

ANALYSIS RESULT

SUSPECTED DEEPFAKE

MEDIUM THREAT

78% CONF

ID: 88345d19-56df-4079-aeba-5c482744cd17 · 12113ms

// AI ANALYST NARRATIVE

THREAT INTELLIGENCE SUMMARY **Detection:** Medium-confidence (78.36%) suspected deepfake identified in a 712x706 PNG image with transparency layer, suggesting potential use of sophisticated image manipulation techniques for synthetic media generation. **Technical Indicators:** The file exhibits characteristics consistent with deepfake imagery, with EXIF metadata stripped (indicating potential operational security measures) and RGBA format supporting overlay techniques commonly used in face-swapping operations. **Recommended Action:** Implement secondary verification protocols for this content and monitor for associated phishing campaigns or disinformation operations, as the medium threat level and alignment with MITRE T1566 (Phishing) and T1565.001 (Data Manipulation) tactics indicate potential use in social engineering or influence operations.

Technical Metadata

Design Decisions & Challenges

AI Models & Why

Claude (Anthropic) — generates structured analyst narratives from detection results; chosen for reliable instruction-following and safe output constraints

Llama via Groq — fast open-source inference for classification and triage tasks; Groq provides ultra-low latency without external API dependency

Prompt Engineering: system prompt enforces clinical language, forbids identity speculation; structured output: Verdict → Indicators → Threat Level → Action
No external detection APIs — all inference runs through Claude & Llama/Groq; no third-party data sharing

User Data & Analysis

Privacy-first: files never stored to disk — processed in memory, discarded immediately

SHA-256 hash only stored for deduplication (not reversible to file contents)

EXIF PII stripped: GPS & device serial removed before storage
IPs hashed before any logging — no raw IP retention

Pipeline: Upload → Hash → Llama/Groq Classification → Metadata Extract → Claude Narrative → Store

Challenges & Solutions

Low detection accuracy initially — single model produced too many false positives; solved by combining Claude + Llama/Groq as a multi-model ensemble for higher confidence verdicts

Outputs were not useful— raw scores were hard to interpret; solved by prompting Claude to generate plain-language explanations pinpointing specific deepfake indicators (e.g. facial boundary artifacts, unnatural blinking)

Users lacked trust in results — binary verdicts felt opaque; solved by surfacing a confidence score alongside a highlighted evidence summary so users can judge reliability themselves

Safety, Privacy & Reliability

What We Store

Data	Stored?	How
File contents	❌ Never	Processed in memory, discarded immediately
Filename	❌ Never	Not accepted or stored
File SHA-256 hash	✅ Yes	For deduplication only, not re-identification
EXIF metadata	Partial	GPS coordinates and device serial numbers stripped before storage
IP addresses	❌ Never raw	Hashed (SHA-256) if logged at all
User identity	❌ N/A	No accounts required in base mode
Submitted URLs	❌ Never raw	Only domain reputation result stored
Analyst narrative	✅ Yes	Contains no PII per system prompt instructions

- We collect no PII (Personal identifiable information)
- We never fetch submitted URLs directly (Google Safe Browsing API)

- Sentinel does not host, cache, or redistribute analyzed media

Checkpoint #4 Roadmap

01 MIT AI Risk Taxonomy Implementation

1. Implement MIT Taxonomy - Use Claude/Groq to classify each result by Entity, Intent, and Timing
2. Tag analysis results with taxonomy categories in JSON response & narrative
3. Dashboard filter by taxonomy category for aggregate threat landscape view
4. Collect user study data and validate whether Sentinel reduces analyst triage time

02 User Study Design

Claim to validate: Sentinel reduces triage time vs. current workflow

Participant:	8-12 students (media/CS background)
Task:	Verify 5 media items w/ & w/o Sentinel
Measure:	Time-to-verdict, confidence, errors
Survey:	NASA-TLX cognitive load + SUS usability
Analysis:	Paired t-test: triage time difference

THANK YOU

Any questions?